Mehmet Selman YILMAZ

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Computer Science and Engineering graduate with experience in applied machine learning and software development at Baykar, KTH, BMC and Intertech. Focused on Python/C++, NLP and recommendation systems, with projects in retrieval, audio processing and real-time applications.

EDUCATION

Sep 2021 - Jun 2026 **B.Sc. in Computer Science & Engineering**, *Sabanci University*, **GPA:** 3.44 / 4.00

Relevant Courses Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, LLMs: Foundations and Practice, Network Science, Introduction to Data Science, Statistical Modelling, Probability, Linear Algebra, Algorithms, Data Structures, Discrete Mathematics, Signals.

WORK EXPERIENCE

Jul - Sep 2026 **NLP and Graph Reasoning Research Intern**, *KTH Royal Institute of Technology*, Stockholm, Sweden

- Investigated how **Llama-3.1-8B** reads and reasons over text-encoded graphs under Prof. Aristides Gionis. A query-local adjacency cue increased balanced accuracy from 80.68% to 98.63% on 1,750 questions across 120 bipartite graphs.
- Ported the Talk Like a Graph benchmark into **Python**/**PyTorch** evaluation pipelines and ran GPU experiments on Berzelius with **Slurm**. The graph-structure experiment evaluated 27,600 responses across seven graph families against algorithmic ground truth.

Aug - Sep 2025 **Artificial Intelligence Software Engineering Intern**, *Baykar*, Istanbul, Türkiye

- Built a recommendation pipeline across Instagram, Twitter and Next Sosyal, combining real/synthetic interactions, **TurkishBERTweet** embeddings, **Two-Tower** retrieval and **Wide & Deep** re-ranking.
- Evaluated recommendations across 5,170 users, comparing Recall@K for the personalized feed with following-based and encoder-only baselines.

Jul - Aug 2025 **Software Engineering Intern**, *BMC*, Izmir, Türkiye

- Built a **NestJS**/**Next.js** AI assistant connected to SAP inventory data, allowing users to retrieve stock information through natural-language queries with the OpenAI API.
- Implemented ABAP material-tracking modules and automated stock-alert workflows to replace manual inventory lookups.

Jun - Aug 2024 **Software Engineering Intern**, *Intertech*, Istanbul, Türkiye

- Developed a financial-insights **RAG** application with **Python**, **SQL**, Node.js and Next.js; automated daily bulletin collection and stored processed content in Azure Blob Storage.
- Deployed the service through a **CI/CD** pipeline using **Docker**, **Kubernetes** and Azure, connecting data ingestion with the application workflow.

Jun 2023 - Jun 2024 **Learning Assistant - Programming Fundamentals**, *Sabanci University*, Istanbul, Türkiye

- Supported **C++** programming instruction for 500+ students and graded assignments and examinations.

PROJECTS

Apr 2026 - Present **adviSU - Academic Advising with RAG** ↗

- Developed a bilingual academic adviser with **FastAPI**, **MongoDB** and **React**. Field-weighted BM25F, multilingual embeddings and reciprocal-rank fusion raised held-out Recall@10 from 59.92% to 94.19% over 30,343 chunks (499 queries).
- Implemented degree audits and conflict-free schedules in code, with evidence-based answers and abstention when data is missing.

PROJECTS (CONTINUED)

PURE Research **SparseViz - C++ Performance Engineering**
- Accelerated sparse-matrix ordering algorithms by 6x in **C++** while contributing to a CPU/GPU performance-profiling and visualization library.

Sep 2025 - Jan 2026 **Music Emotion Recognition on DEAM**
- Built a segment-CNN audio pipeline with attention pooling and training-only normalization; achieved test MSE 0.0114, Pearson correlation 0.721 and CCC 0.689.

Sep 2025 - Present **EchoSpaceAR - AR Sound Awareness** ↗
- Connected a **Unity**/**C#** AR interface to YAMNet sound classification and Whisper speech recognition, translating stereo audio into real-time visual cues.

Sep 2024 - Jan 2025 **Intent Detection and Slot Filling**
- Reproduced **Slot-Gated BiLSTM** on SNIPS for joint intent detection and slot filling; compared **LSTM**/BiLSTM and classical classifiers with FastText embeddings and Bayesian hyperparameter search.

Apr 2026 - Present **AutoBEME - Market-Based Ensemble Learning** ↗
- Built a **scikit-learn**-compatible market-based ensemble; beat SVM baselines in two of three benchmark regimes, including a Macro-F1 increase from 0.826 to 0.864.

Oct 2026 - Present **Atcha - Shared Questions, Games and Memories**
- Developed a **Flutter**/**Supabase** application with real-time shared state, offline retries and on-device multilingual **Sentence-BERT** inference.
- Applied **PostgreSQL** row-level security and idempotent writes; optimized a measured history query from about 500-535 ms to 180-205 ms after deployment.

2026 **BlueSky - Real-Time Feed Generator** ↗
- Built real-time custom feeds from the AT Protocol firehose using BERTurk embeddings and topic classification, with a Flask/**PostgreSQL** stack deployed on Railway.

SKILLS

Languages Turkish (Native), English (Advanced)

Programming **Python**, **C++**, **C#**, **SQL**; algorithms, data structures, object-oriented programming

Machine learning **PyTorch**, **TensorFlow**, **scikit-learn**; CNNs, **LSTM**/BiLSTM, **Transformers**, **BERT**, **Llama-3.1-8B**, RAG, embeddings, recommendation, model evaluation

Systems Git, **Docker**, **Kubernetes**, **Azure**, **CI/CD**, **Slurm**, **FastAPI**, **MongoDB**, PostgreSQL

Integration and testing Real-time audio/stream processing, **Unity**; unit, integration, E2E tests

ACTIVITIES

Jun 2023 - Apr 2025 **Vice President, IEEE Computer Society, Sabanci University.** Co-organized CompTalks and an algorithm competition.

VOLUNTEER WORK

Feb - Jun 2022 **Sabanci University Civic Involvement Projects.** Volunteered in weekly educational activities with elementary school students.

INTERESTS

Table tennis, cinema, travel, chess and billiards.